

Course Code : ECT 457-1

MNQR/RS – 23 / 1096

**Eighth Semester B. E. (Electronics and Communication Engineering)
Examination**

**REAL TIME OPERATING SYSTEMS
AND KERNELS**

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as mentioned.
 - (2) Draw neat and clean diagram, wherever necessary.
 - (3) Use of Non – programmable calculator is permitted.
 - (4) Assume suitable data wherever necessary.
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1. (a) The resource utilization shows up different forms in different types of operating systems. Illustrate what resources must be managed carefully in the following settings :
 - (a) Mainframe or minicomputer systems.
 - (b) Workstations connected to servers.
 - (c) Mobile computers. 5(CO1)
 - (b) What are differences between user-level threads and kernel-level threads ? Under what circumstances is one type better than the other ? Justify your answer. 5(CO1)
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2. (a) What is the main difficulty that a programmer must overcome in writing an operating system for a real-time environment ? 5(CO2)

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- (b) Consider the following snapshot of a system :

	<u>Allocation</u>				<u>Max</u>				<u>Available</u>			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2	1	5	2	0
P_1	1	0	0	0	1	7	5	0				
P_2	1	3	5	4	2	3	5	6				
P_3	0	6	3	2	0	6	5	2				
P_4	0	0	1	4	0	6	5	6				

Answer the following questions using the Banker's algorithm :

- (a) What is the content of the matrix Need ?
 - (b) Is the system in a safe state ?
 - (c) If a request from process P_1 arrives for (0, 4, 2, 0), can the request be granted immediately ? 5(CO2)
3. (a) Using the program shown in Figure, find the output that will be generated at LINE A :

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int value = 5;

int main()
{
    pid_t pid;

    pid = fork( );

    if (pid == 0) { /* child process */
        value += 15;
        return 0;
    }
    else if (pid > 0) { /* parent process */
        wait(NULL);
        printf("PARENT: value = %d", value); /* LINE A */
        return 0;
    }
}
```

5(CO3)

- (b) Disabling interrupts frequently can affect the system's clock. Explain why this can occur and how such effects can be minimized ? 5(CO3)
4. (a) The Linux kernel does not allow paging out of kernel memory. What effect does this restriction have on the kernel's design ? What are two advantages and two disadvantages of this design decision ? 5(CO4)
- (b) Why Windows, Linux and Solaris implement multiple locking mechanisms ? Demonstrate the circumstances under which they use spin-locks, mutex locks, semaphores, adaptive mutex locks and condition variables. In each case, explain why the mechanism is needed. 5(CO4)
5. Consider a file currently consisting of 100 blocks. Assume that the file-control block (and the index block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked and indexed (single-level) allocation strategies, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory :
- (a) The block is added at the beginning.
- (b) The block is added in the middle.
- (c) The block is added at the end.
- (d) The block is removed from the beginning.
- (e) The block is removed from the middle.
- (f) The block is removed from the end. 10(CO4)
6. Dynamically loadable kernel modules give flexibility when drivers are added to a system. But do they have disadvantages too ? Under what circumstances would a kernel be compiled into a single binary file, and when would it be better to keep it split into modules ? Explain your answer. 10(CO5)

